Faeze Mohammadi

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Research Profile

Computer Engineering undergraduate at the University of Tehran interested in machine learning and deep learning, especially how data and learned representations shape model behavior. I work with Dr. Ahmad Kalhor in the Machine Learning Laboratory on evaluating data and representations with the Separation Index, and with Malihe Khatami in Dr. Siamak Mohammadi's Dependable System Design Laboratory on neural networks for side-channel analysis. Across my work I rely on controlled experiments, matched baselines, multi-seed runs, and honest reporting of negative results.

Research interests: Machine learning · Deep learning · Computer vision · Cryptography

Education

University of Tehran — BSc in Computer Engineering Sep 2024 – present (expected Jul 2028)
GPA: 19.68/20. Ranked **1st of ~120** in the Computer Engineering entering cohort (first two years).
Selected grades (20/20): Probability and Statistics; Discrete Mathematics; Artificial Intelligence; Data Structures; Formal Languages and Automata; Engineering Mathematics; Advanced Programming; Calculus II; Differential Equations.
Selected current coursework (Fall 2026): Machine Learning; Signals and Systems.

National Organization for Development of Exceptional Talents — Secondary education 2018 – 2024
Six years of secondary education. **Diploma GPA: 19.86/20.**

Research Experience

Evaluating Data and Neural Representations with SI and SmI Oct 2026 – present
Machine Learning Laboratory, University of Tehran | Supervisor: Dr. Ahmad Kalhor

- Studying how the Separation Index (SI) and Smoothness Index (SmI) can assess data difficulty and diversity, track how representations evolve across layers during training, and guide feature or layer reduction without loss of accuracy; the project is in its first phase.
- Exploratory first phase: compared sample-selection methods on MNIST, Fashion-MNIST, CIFAR-10, and CIFAR-100 using accuracy, SI, geometric coverage, and density, with separate training and validation sets.
- Now extending to the mean and variance of SI across data subsets as a diversity criterion, followed by SmI and the layer-wise analysis.

Neural Network-Based Side-Channel Attack Analysis for Cryptographic Algorithms Oct 2026 – present
Dependable System Design Lab, University of Tehran | Supervisor: Malihe Khatami (PhD candidate) | PI: Dr. Siamak Mohammadi

- Part of the laboratory's project *Design and Implementation of Remote Side-Channel Attacks and Countermeasures*.
- Benchmarked MLP, CNN, LSTM, and a CNN with a linear-attention adapter on the public ASCAD dataset of AES power traces (50,000 profiling / 10,000 attack traces) to recover a single AES key byte.
- CNN, MLP, and the hybrid model recovered the target byte across all three seeds; LSTM models did not, and these negative results are documented.
- Ran attention ablations, checkpoint replay, and multi-seed key-rank analysis; the hybrid's recovery efficiency was mixed relative to matched CNN controls.

Teaching Experience

Teaching Assistant and TA Supervisor — Discrete Mathematics Fall 2025 – present (3 terms)
University of Tehran | Instructor: Dr. Siamak Mohammadi

- Help design homework assignments and quizzes and lead small-group problem-solving sessions in a course of 90+ students; supervising the TA team this term.

Teaching Assistant — Formal Languages and Automata Fall 2026 – present
University of Tehran | Instructor: Dr. Hossein Hojjat

- Supporting students with formal languages, grammars, finite automata, and formal proofs.

Selected Projects

Persian Poetry Authorship Attribution and Retrieval
Repository | Python, scikit-learn, Tesseract, Vosk

- Built attribution and retrieval benchmarks on 500 poems by five poets (300/100/100 split) with duplicate controls and author-name masking; character TF-IDF + LinearSVC reached 0.95 macro-F1 on held-out poems.
- Added OCR and speech input in a local demo, evaluated separately on a small pilot.

Separation Index Representation Audit
Repository | Independent project | Python, PyTorch

- Re-implemented a published SI definition on Optdigits MLPs and Fashion-MNIST CNNs with paired seeds and fixed baselines; compared SI with held-out accuracy and reported the cases where it did not track performance.

Blood-Cell Classification: MLP vs CNN
Repository | Python, PyTorch

- Compared MLP and CNN at matched ~3.18M-parameter budgets; analyzed confusion matrices, confidence, and feature maps, and tested robustness to rotation, translation, and occlusion.

Q-Learning for MountainCar
Repository | Python, NumPy, Gym

- Implemented tabular Q-learning from scratch; compared exploration schedules and reward shaping, and detected reward hacking by measuring goal success separately from training reward.

Other projects: decision trees and softmax regression from scratch · genetic-algorithm melody reconstruction and Minimax Checkers · Bayesian robot localization and Naive Bayes spam filtering (R) · BFS/DFS/IDS/A* search.

Honors

- Ranked 1st of ~120 in the Computer Engineering entering cohort, University of Tehran.
- National Olympiad Round 2 qualifier in Mathematics, Informatics, and Economics & Management.

Skills

Programming: Python, C++, C, R.
Machine learning and data: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, NumPy, pandas.
Languages: Persian (native); English (upper-intermediate, CEFR B2).

Selected Courses and Certificates

Machine Learning Specialization (DeepLearning.AI / Stanford) · CS50 Introduction to AI with Python (Harvard) · Cryptography I (Stanford) · Writing in the Sciences (Stanford) · OpenCV University bootcamps: PyTorch, TensorFlow-Keras, OpenCV, Vision-Language Models · View all credentials

Academic Writing

A Study of the Higgs Boson Particle. Sole-authored literature review (high-school research); manuscript available upon request.

References

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